

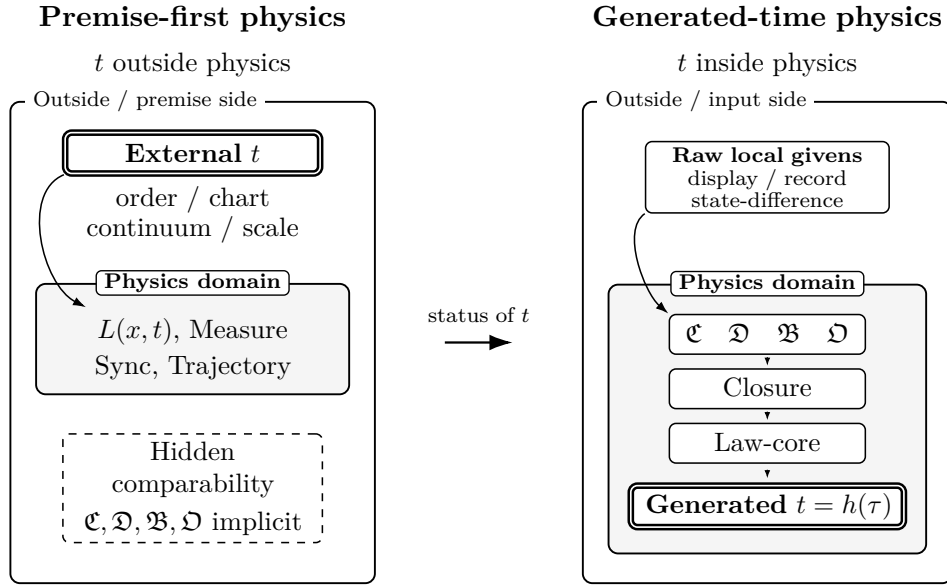


Figure 1: Contrast between the worldview of conventional physics and that developed in this paper. In the conventional scheme, the time variable, continuous time, coordinate time, and frequency are placed among the premises of physical-law formulation. In the scheme developed here, the conditions for establishing a time reference are constructed from local displays, records, object-state differences, and information-coupling positions, and the conventional time variable appears as a representation of the constructed time reference rather than as a premise of law formulation.

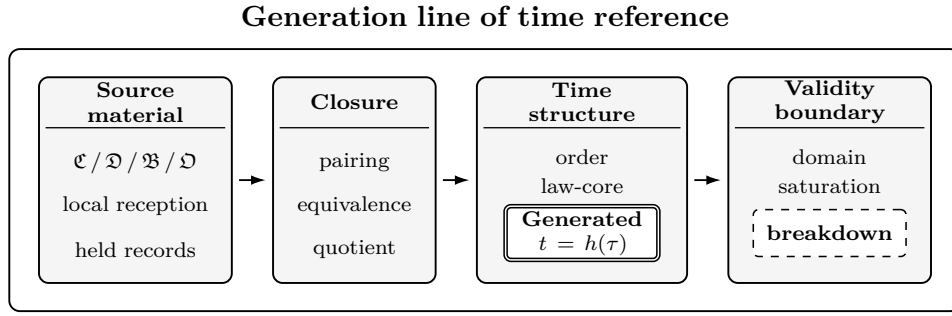


Figure 2: Generation pathway for time reference. The diagram shows the order of construction from local displays, records, object-state differences, and information-coupling positions through comparison closure and time representation to boundary and failure determination.

### Base role types and local operational components

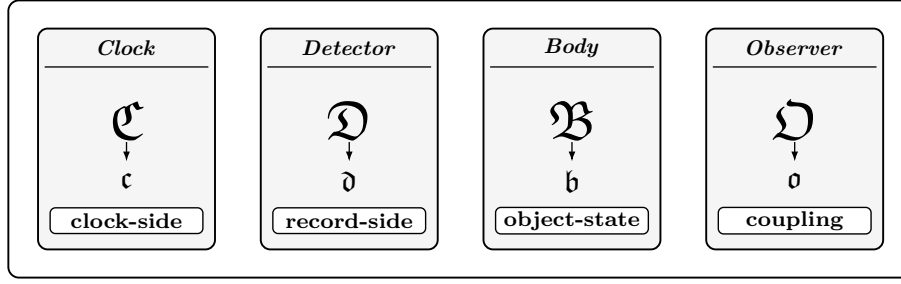


Figure 3: Construction of the base role-type set and the local four-role binding. The four base role types for the clock side, record side, object-state side, and observer-side information coupling are denoted by  $\mathfrak{C}$ ,  $\mathfrak{D}$ ,  $\mathfrak{B}$ ,  $\mathfrak{O}$ . A quadruple of local operational components satisfying these role specifications within the same local scene forms a local four-role binding witness.

### Role instances and physical-layer relations

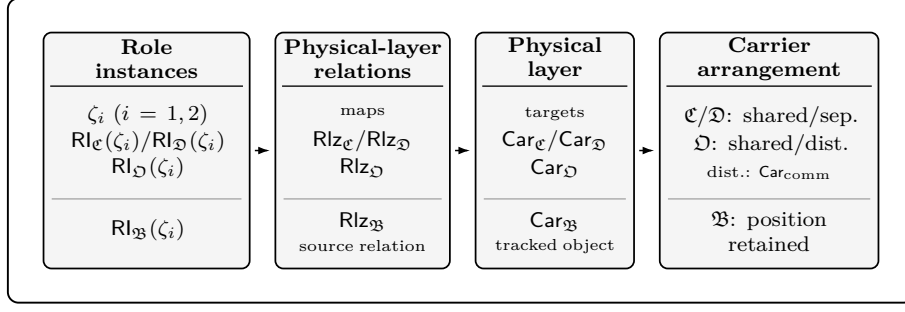


Figure 4: Role instances and physical-carrier realization. Role instances are distinguished for two object-specification indices. The physical-carrier realization maps then assign the  $\mathfrak{C}$ -derived,  $\mathfrak{D}$ -derived, and  $\mathfrak{D}$ -derived role instances to physical carriers, while the  $\mathfrak{B}$ -derived object-state-side instances are associated with the object-state-side entities corresponding to their object specifications. Sharing by the same physical carrier, separation by physical carriers distinct for each object, and distribution across multiple information-coupling carriers are distinguished on the input-formation side by equality relations among the values of the  $\mathfrak{C}/\mathfrak{D}/\mathfrak{D}$ -derived physical-carrier realization maps and by physical connection carriers belonging to  $Car_{comm}$ .

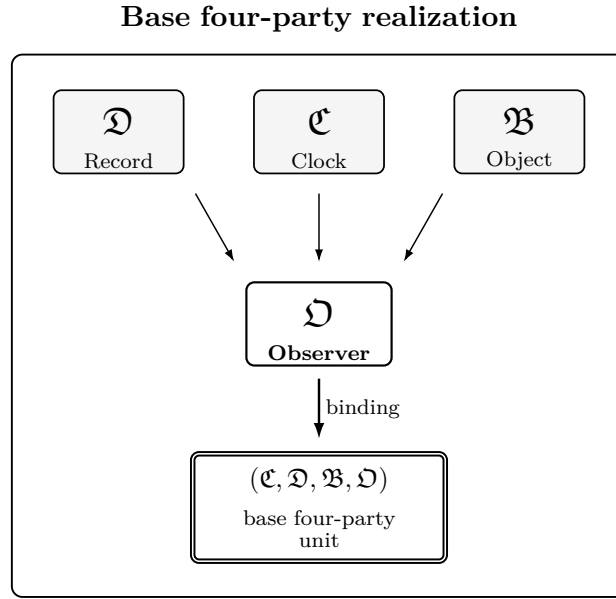


Figure 5: Base four-party realization. When the clock side  $\mathfrak{C}$ , record side  $\mathfrak{D}$ , object-state side  $\mathfrak{B}$ , and observer-side information-coupling position  $\mathfrak{O}$  are given in the same scene, they constitute the base four-party unit of the local time-generation task.

### Comparison quotient generation

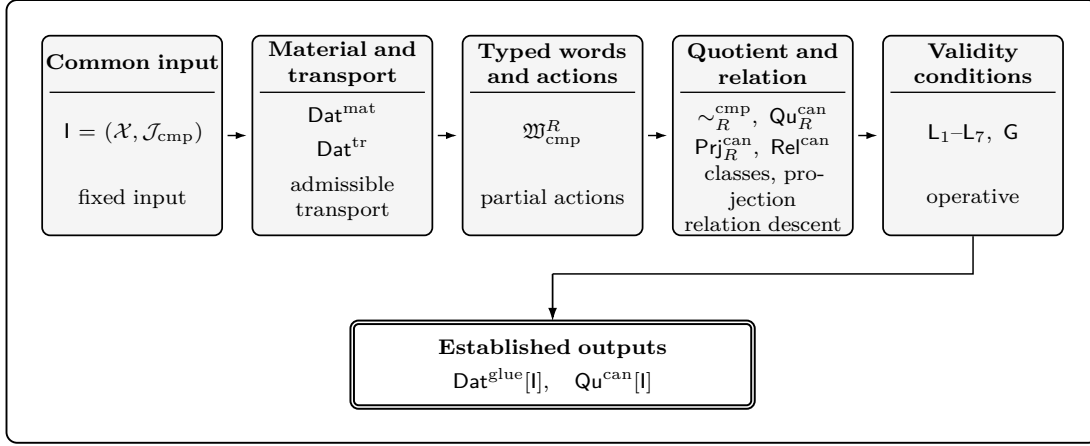


Figure 6: Comparison-stage dependency structure from the comparison-rule-dependent common input to the comparison-gluing datum and the canonical comparison quotient. From the comparison-stage input, comparison-admissible inputs, comparison-transport components, typed finite words, comparison-reachability equivalence relations, component quotients, canonical projections, and comparison two-role relations on the quotients are determined in dependency order. The seven comparison-stage conditions and the condition on all pure-transport closed loops determine whether each construction stage is established.

**Recursive composition from local  
four-role binding to common input**

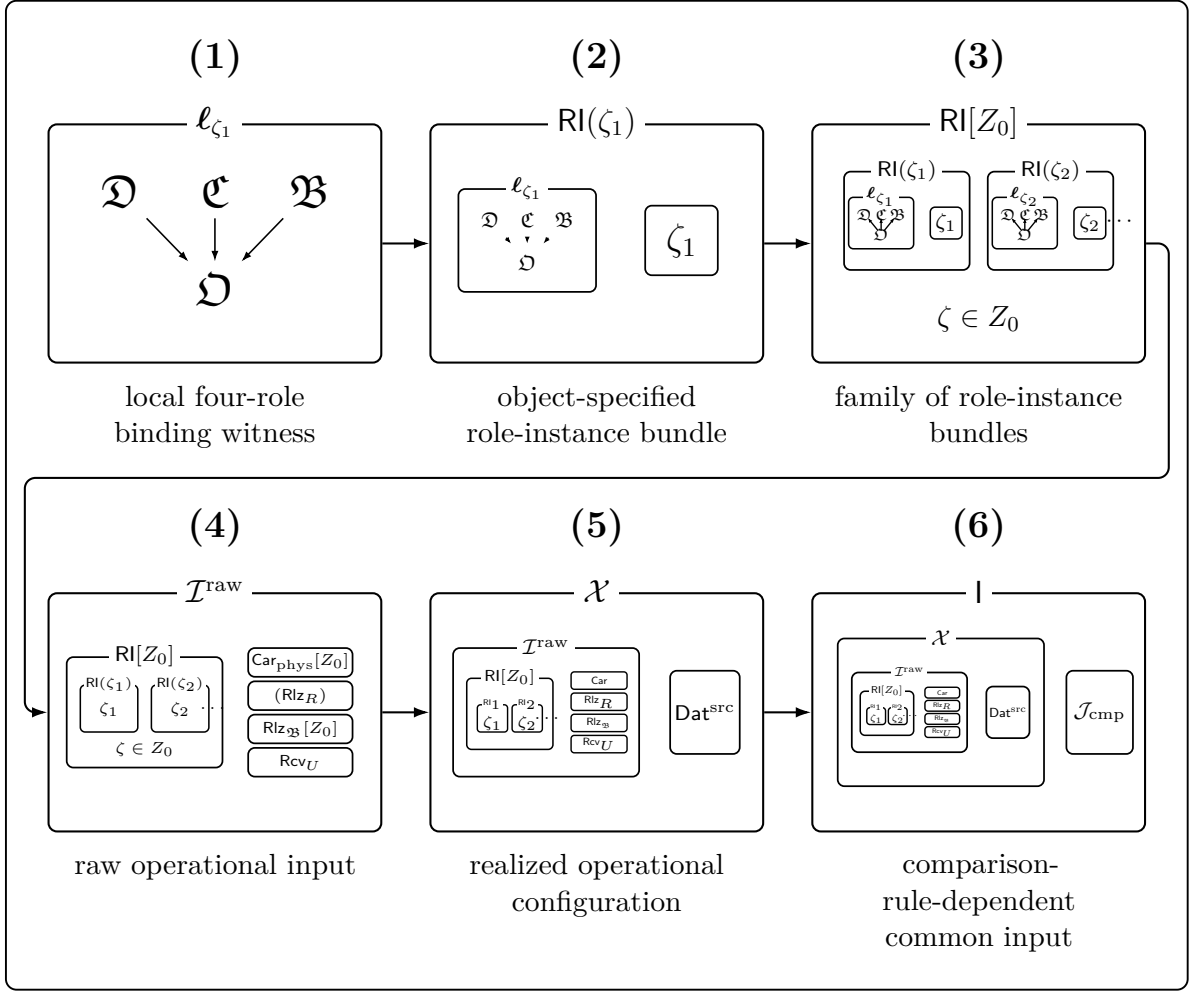


Figure 7: Recursive composition from the local four-role binding to the comparison-rule-dependent common input. An object defined as a whole at one stage is retained as a structured component at the next stage. The construction proceeds from the local four-role binding witness  $\ell_{\zeta_1}$ , through the object-specified role-instance bundle  $\text{RI}(\zeta_1)$ , the family  $\text{RI}[Z_0]$ , the raw operational input  $\mathcal{I}^{\text{raw}}$ , and the realized operational configuration  $\mathcal{X}$ , to the comparison-rule-dependent common input  $\mathcal{I} = (\mathcal{X}, \mathcal{J}_{\text{cmp}})$ .

### Comparison reachability and equivalence classes

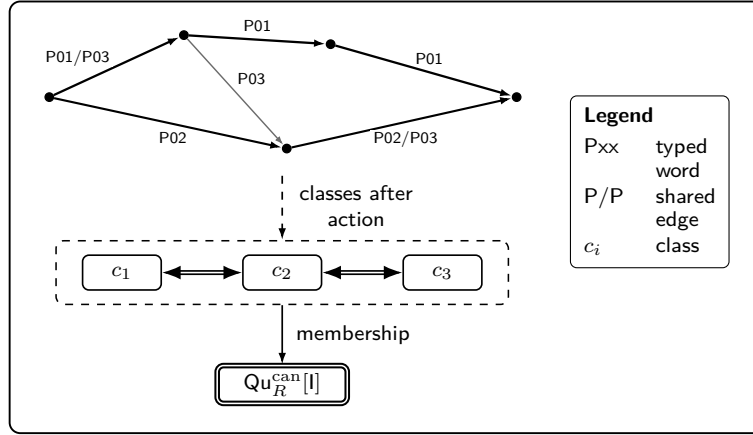


Figure 8: Actions of extended comparison words and comparison-reachability classes. Assume  $L_{\leq 2}[Ev^{cmp}]$  and fix  $R \in Rol^{cmp}$ . The dots represent reception-phase types and the edges represent admissible atomic actions.  $P01$ ,  $P02$ , and  $P03$  denote three extended comparison words  $e_1, e_2, e_3$  with the same source and target types. Take a point  $x$  in the common domain of the three actions and set  $y_i := Act_R^{ext}[l][e_i](x)$  and  $c_i := Proj_R^{can}[l](y_i)$ . Definition 4.25 and Definition 4.68 give  $x \sim_R^{cmp}[l]y_i$ . Lemma 4.71 then gives  $c_1 = c_2 = c_3 = Proj_R^{can}[l](x)$ . The lower part shows that this common class belongs to  $Qu_R^{can}[l]$ .

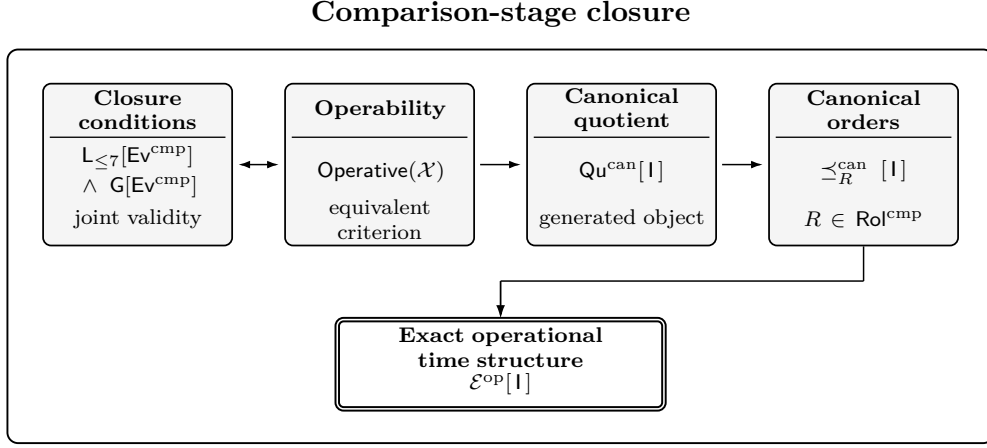


Figure 9: Separation between comparison-stage closure and the subsequent object sequence. Comparison-stage closure yields the operability judgment; the canonical comparison quotient derived from the C and D roles, the descent of canonical order data, and the exact operational time structure are determined as subsequent objects.

### Representational coordinate atlas

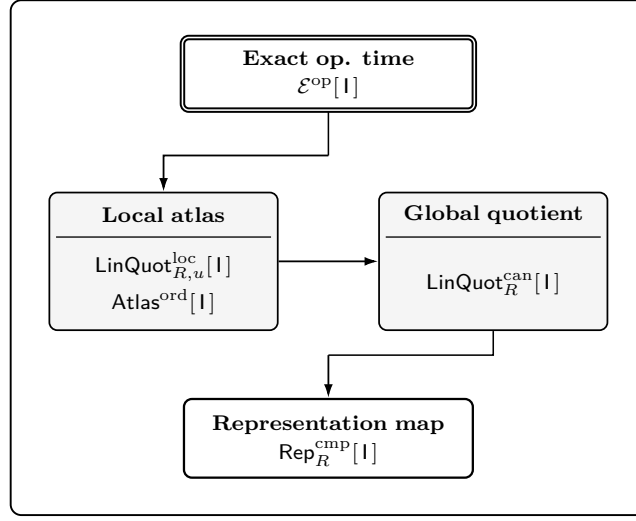


Figure 10: Generation order under  $\text{ReprCtx}_{\leq 3}[I]$ , from exact operational time structures through local order coordinates and the comparison-order atlas to global order-coordinate quotients. The final comparison representational coordinate is the composition of the order embedding into the selected representational value range with the canonical global order-coordinate projection.

### Dynamics-level generation

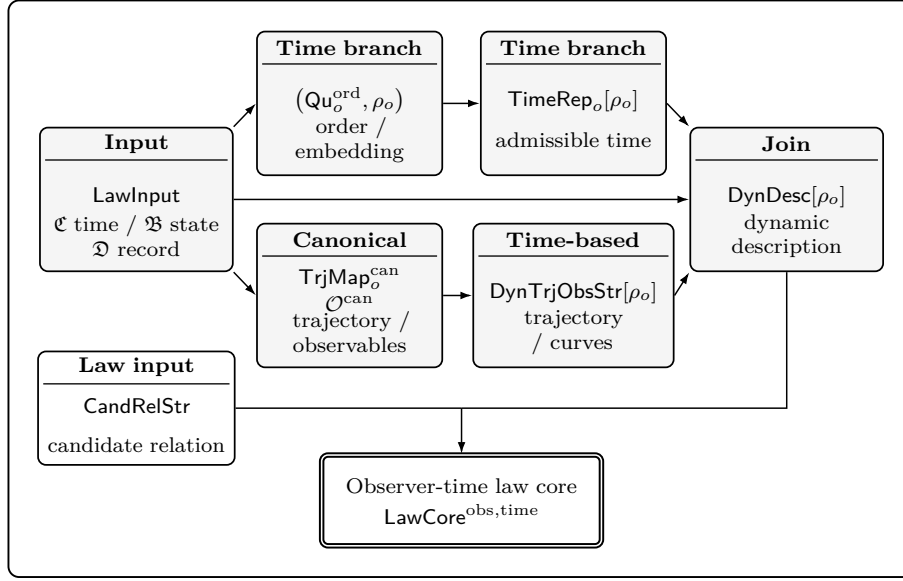


Figure 11: Generation at the dynamical-description stage. Under the conditions for each stage, the observer-time law-input datum yields the time structure and the trajectory-and-observation structure, which form the dynamical-description object. This object and the law-candidate relation structure determine the observer-time law core.

### Operational input formation

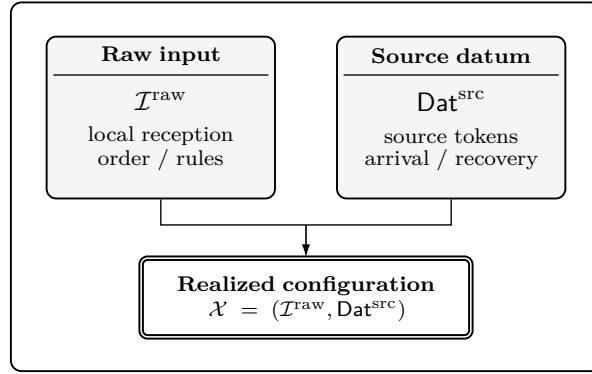


Figure 12: Formation of operational inputs. Local displays, records, object-state differences, and observer-side coupling materials are assembled as a realized operational configuration passed to the comparison and dynamical-description stages.

### Maximal effective interval and first excess

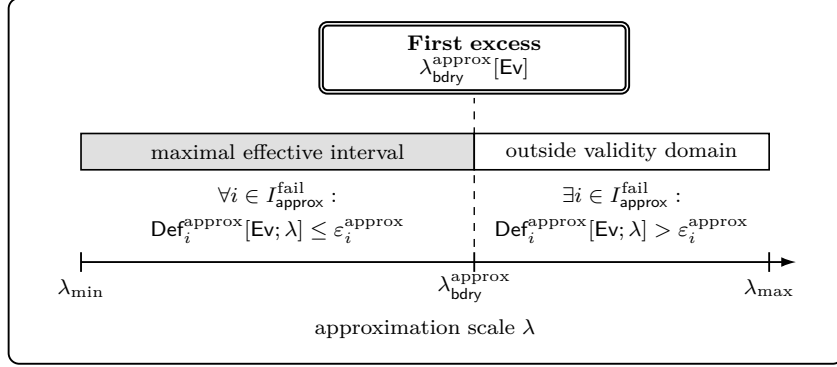


Figure 13: The maximal effective interval and first excess for continuum approximation. When the inputs for quantitative evaluation are satisfied at every scale, the nine-component defect-quantity family is monotonically nondecreasing with respect to the approximation scale, and the boundary scale is defined, the maximal effective interval equals the scale validity domain. Every scale below the boundary scale belongs to the maximal effective interval; the boundary scale lies outside the scale validity domain and is the least approximation scale at which an excess component first occurs.

### Validity and failure localization

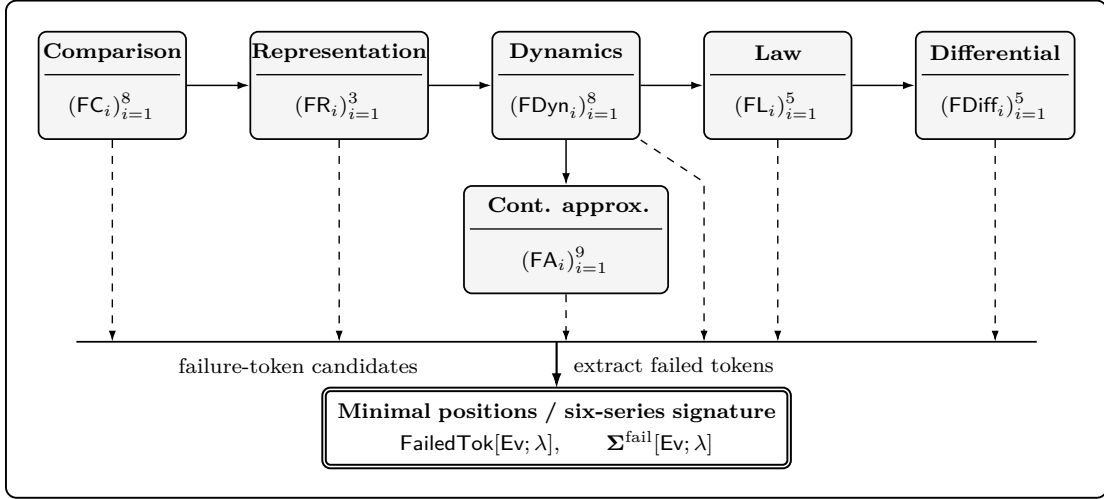


Figure 14: Six-group localization of validity and failure. The minimal failure-position set is defined as the set of positions minimal with respect to the failure-generation dependency order among the failure-position tokens belonging to the comparison, representation, continuum-approximation, dynamical-description, law-representation, and differential-consistency and law-form-preservation stages.

### Five-valued audit report generation

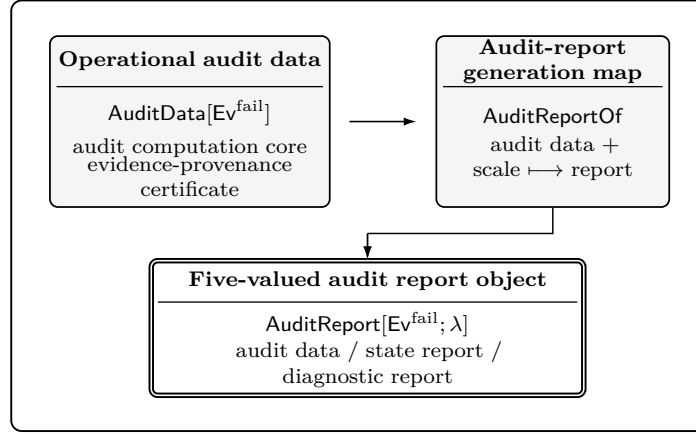


Figure 15: Map generating a five-valued audit-report object from operational audit data. The audit data, comprising the audit computation core and evidence-provenance certificate, are supplied together with the scale to the audit-report generation map **AuditReportOf**, which sends them to **AuditReport**, whose components are the audit data, audit-state report, and audit-diagnostic report.

## Finite audit procedure

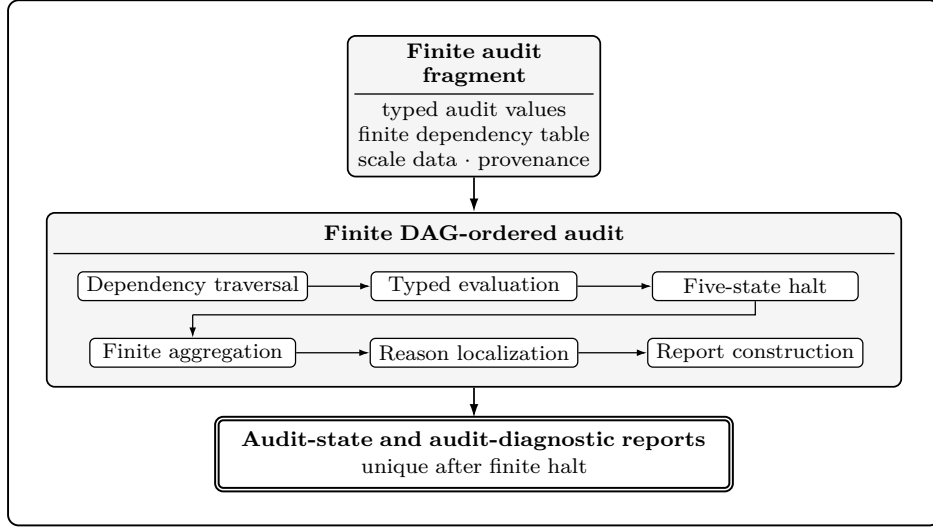


Figure 16: Finite procedure generating audit-state and audit-diagnostic reports from a finite audit fragment. It receives typed audit values, a finite dependency table, scale and boundary data, and finite evidence-provenance records; it executes dependency-order traversal, typed condition evaluation, five-valued halting, finite aggregation, reason localization, and report generation in finite-DAG order. If the operational audit datum is also supplied, its tuple with the two resulting reports gives **AuditReport**.

### Six-family localization of engineering evidence

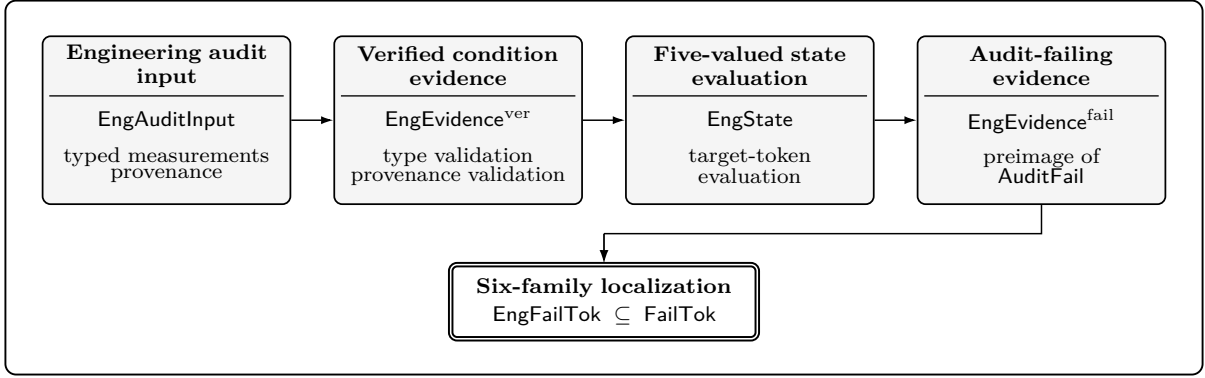


Figure 17: Localization of engineering evidence within the six series under a fixed scale-augmented complete evaluation specification  $(\mathbf{Ev}, \lambda)$ . Common arguments of the objects in the figure are omitted. Verified engineering formal-condition evidence is formed from engineering audit input, the five-valued audit state is evaluated, and audit-failing evidence is extracted. Its target image  $\mathbf{EngFailTok}[\mathbf{Ev}; \lambda]$  is localized as a subset of **FailTok**.

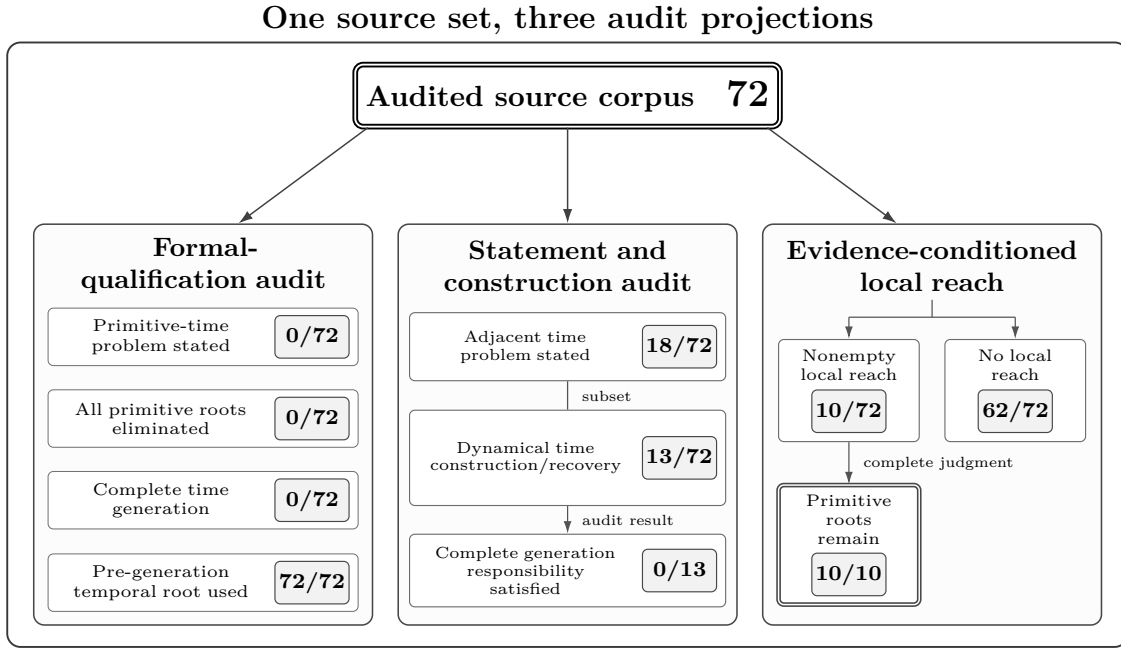


Figure 18: Three audit surfaces for the same 72 references: formal qualification, statements and constructions, and local reachability conditional on the evidence environment. Subsets defined by statements and constructions and local reachability conditional on the evidence environment are separated as distinct axes of determination.

### Canonical quotient cascade to time and trajectory representations

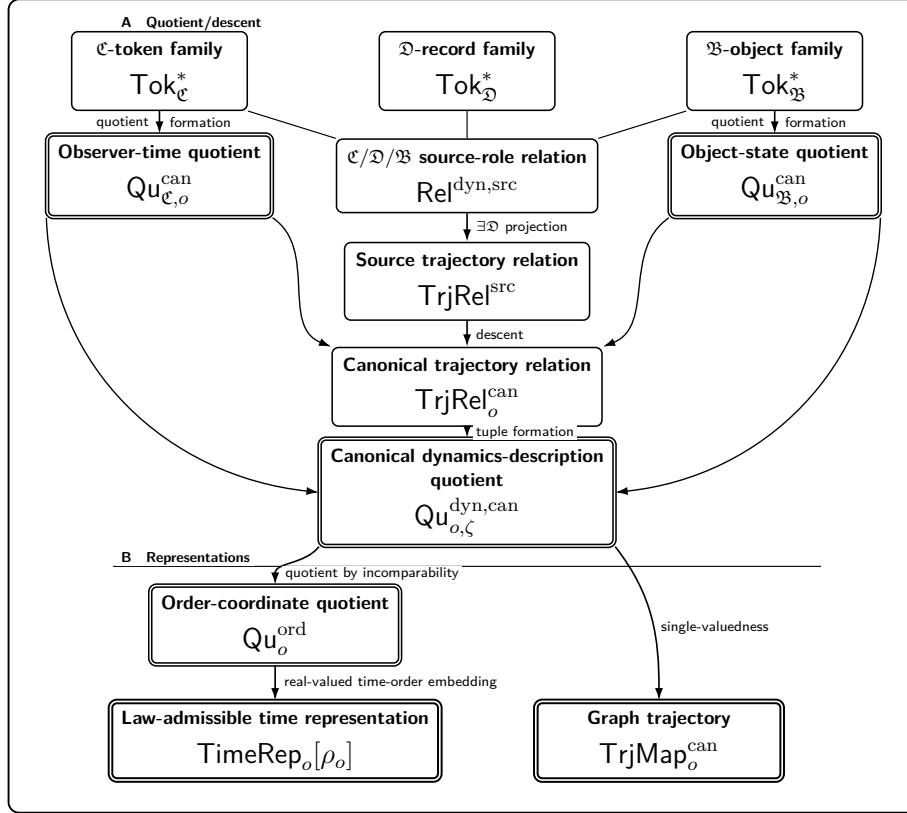


Figure 19: Dependency structure from the canonical quotient cascade to time representations and trajectory maps. The upper row constructs the canonical dynamical-description quotient from the token families derived from  $\mathcal{C}$ ,  $\mathcal{D}$ , and  $\mathcal{B}$ , through the observer-time quotient, object-state quotient, and canonical trajectory relation. The lower row distinguishes the branch that obtains a law-admissible time representation from the time-order side through the quotient by incomparability and a real-valued time-order embedding, from the branch that obtains a graph trajectory from the single-valuedness of the canonical trajectory relation.

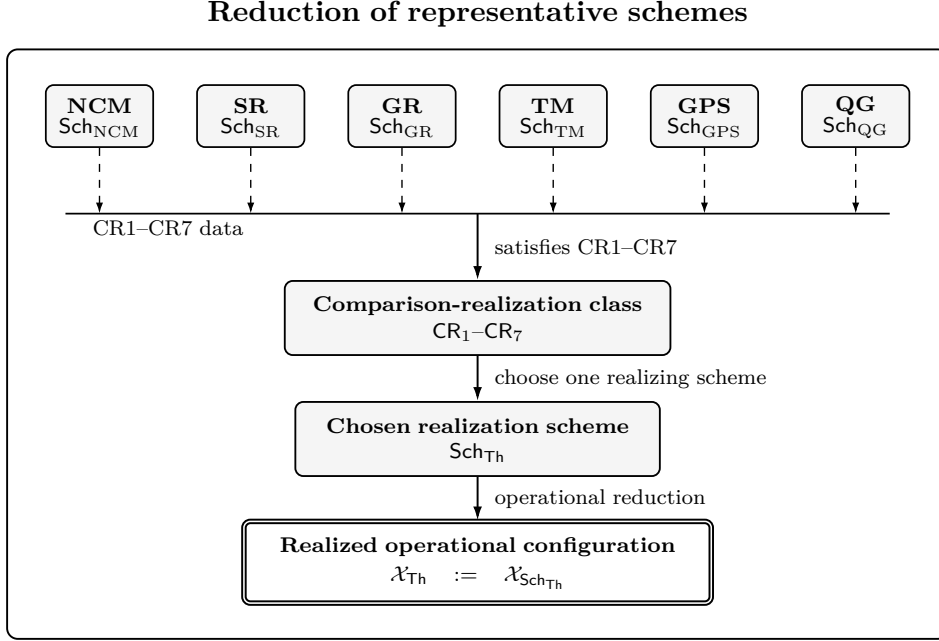


Figure 20: Reduction from the six representative schemes to theory-specific realized operational configurations. The theory corresponding to each representative scheme belongs to the comparison-realization class by supplying the data from CR1 through CR7. Once a realization scheme  $Sch_{Th}$  is selected for a theory  $Th$ , the operational reduction defines  $\mathcal{X}_{Th} := \mathcal{X}_{Sch_{Th}}$ .

## Two independent inheritance audits

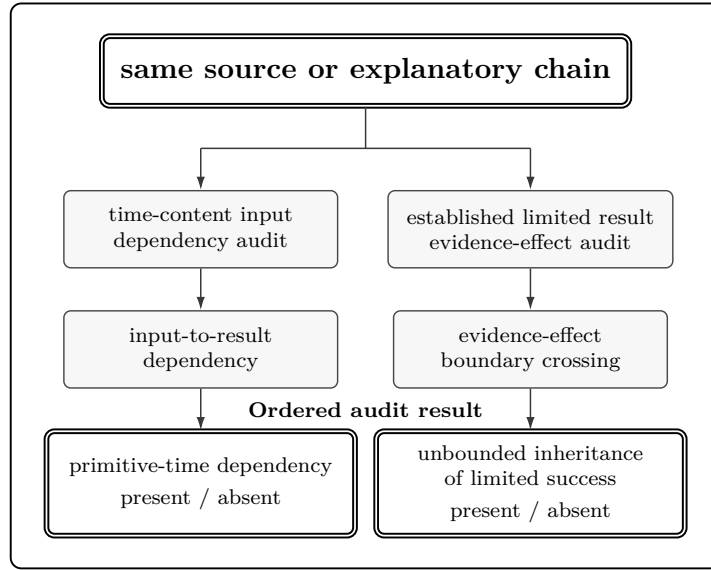


Figure 21: Primitive-time-dependence result inheritance and transgression of the evidence-effect domain of limited results are determined as distinct audit axes. The two determinations are not identified and are retained as an ordered pair for the same reference or explanatory sequence.

The target changes while the evidence is carried forward

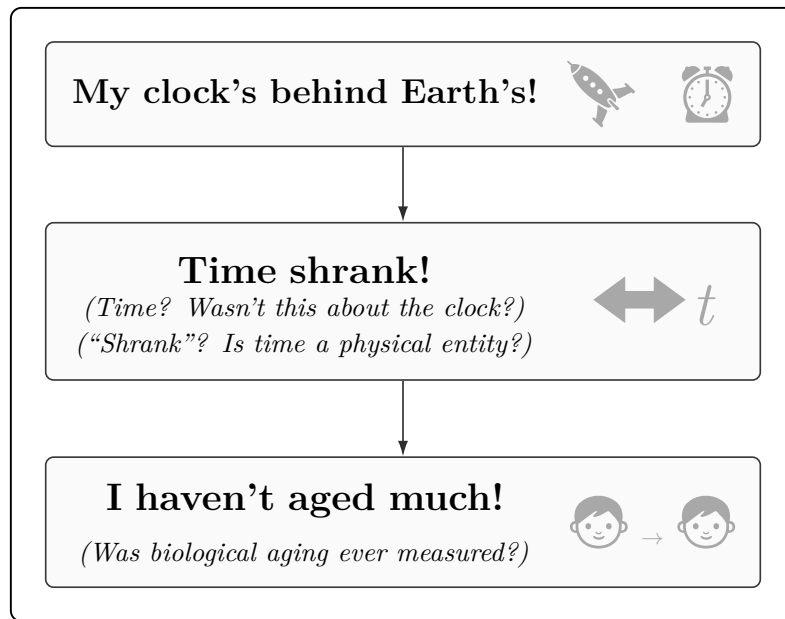


Figure 22: The object under discussion shifts among a clock, time, and a biological process; two questions identify the evidence boundary at each transition.

## Target-consistency audit of statements about time

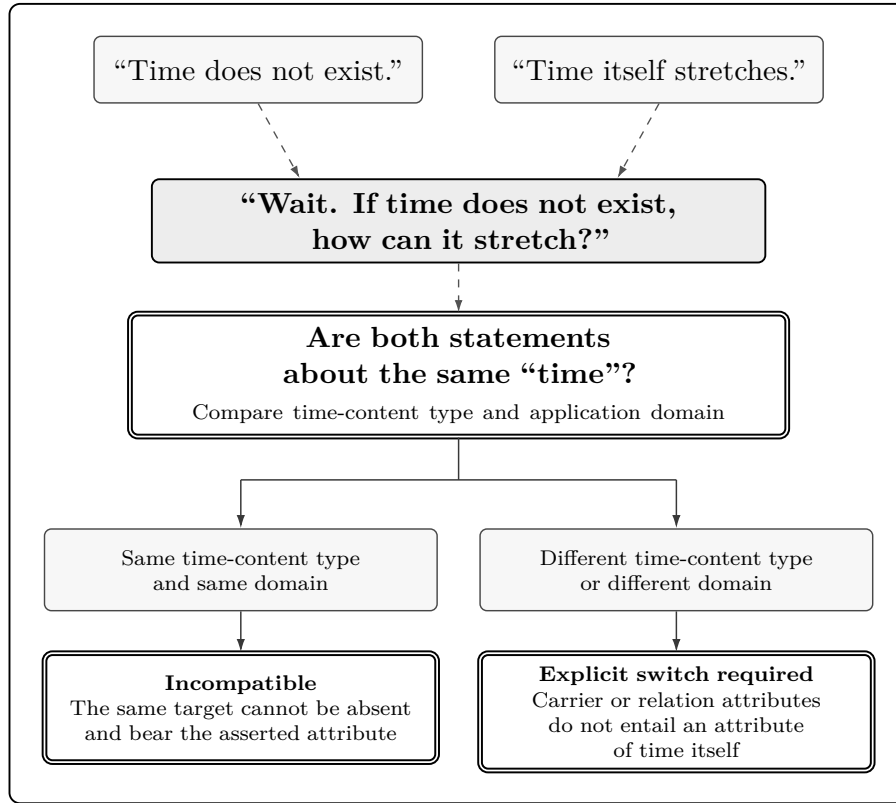


Figure 23: Target consistency between time-nonexistence statements and time-property statements. The upper row gives representative forms of the two kinds of statement subjected to audit, while the gray band gives the question that arises for a reader who encounters them in succession. These are not verbatim quotations from an original source. The dashed lines from the upper row to the central audit box indicate reading order and do not express formal implication. Solid lines from the central box onward indicate the target-consistency determination. The two statements are incompatible when they concern the same type and the same domain. Different types or domains require an explicit change, and a property of a physical carrier or relational quantity does not entail a property of time itself independently of physical carriers.

## Evaluation success and qualification consequences

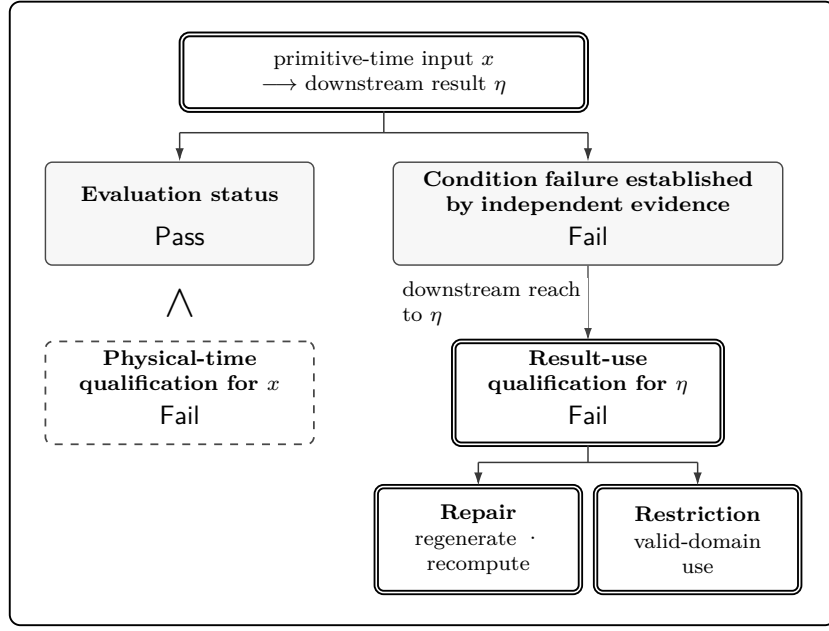


Figure 24: Coexistence of evaluation success and two qualification failures. The left branch shows evaluation success coexisting with the failure of physical-time qualification for a primitive-time input. The right branch shows failure of a generation or establishment condition, established by independent evidence, reaching the qualification for use of downstream results and requiring repair or use within the domain of validity.

## Result-preserving re-grounding

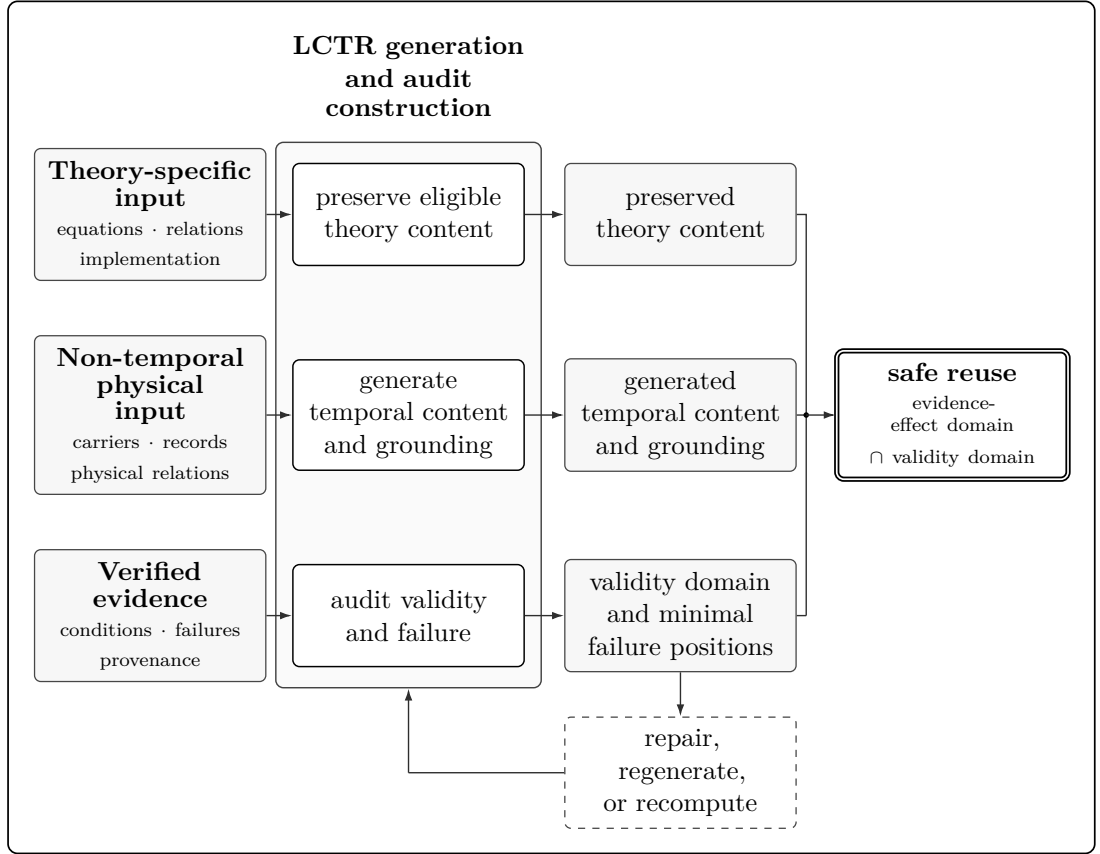


Figure 25: Result-preserving re-grounding in Definition 8.99. The retention pathway for existing theoretical content is separated from the generation-and-audit pathway for temporal qualification, and the safe domain of reuse is determined as the intersection of the evidence-effect domain and the validity domain. After invalidation, the construction is re-audited through repair, regeneration, and recomputation.

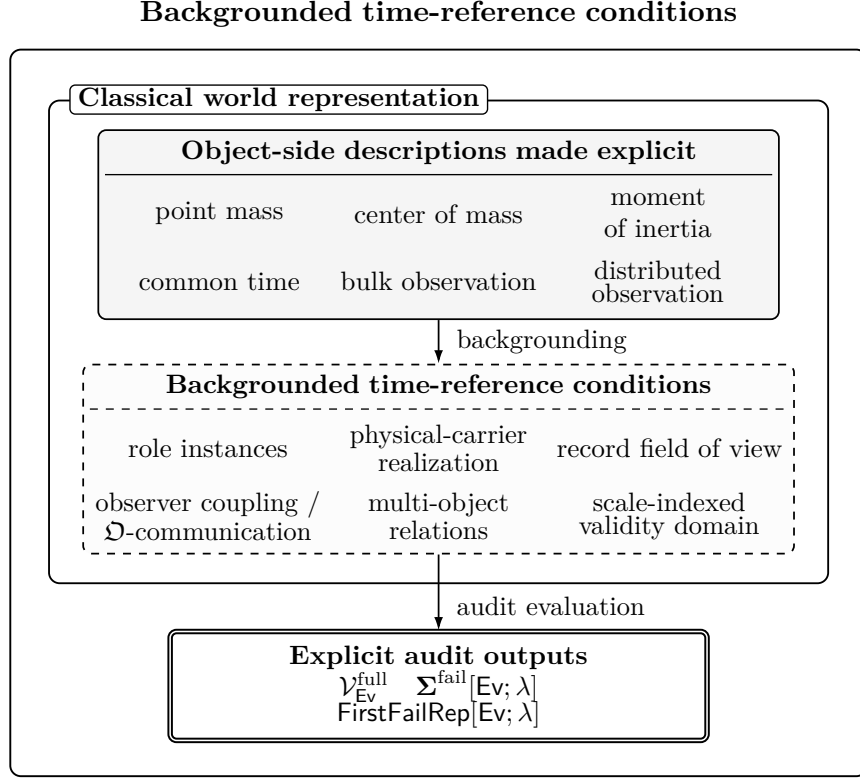


Figure 26: Object-side physical quantities made explicit by a classical world representation and the establishment conditions for time reference that it backgrounds outside the expressions. Multi-object descriptions using point masses, centers of mass, moments of inertia, and common time correspond to role instances, physical-carrier realizations,  $\mathfrak{D}$ -derived record field of view,  $\mathfrak{D}$ -communication, multi-object relations, scale-indexed validity domains, six-series invalidation signatures, and audit reports.

# **Bulk-record / following- record audit comparison**

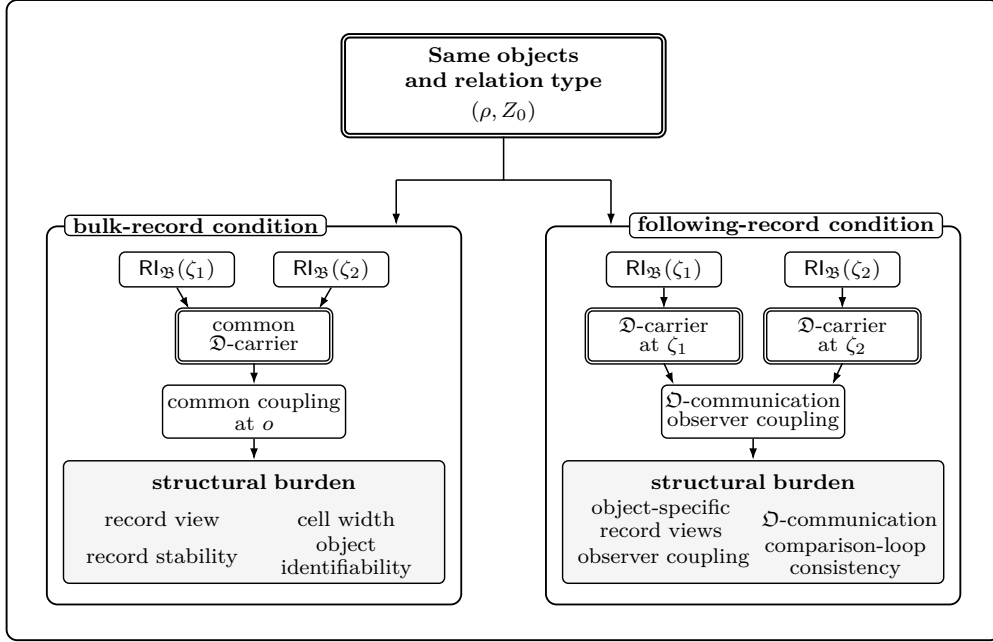


Figure 27: Structural burdens relative to engineering evidence for the bulk-recording and following-recording conditions. For the same object-specification set and relation index, the bulk-recording condition requires record field of view, cell width, record stability, and object identifiability in a single  $\mathfrak{D}$ -carrier. The following-recording condition requires  $\mathfrak{D}$ -communication, observer-side coupling, and comparison-loop consistency in addition to object-specific record fields of view in distinct  $\mathfrak{D}$ -carriers.

### Configuration-specific outputs of the classical multi-object audit

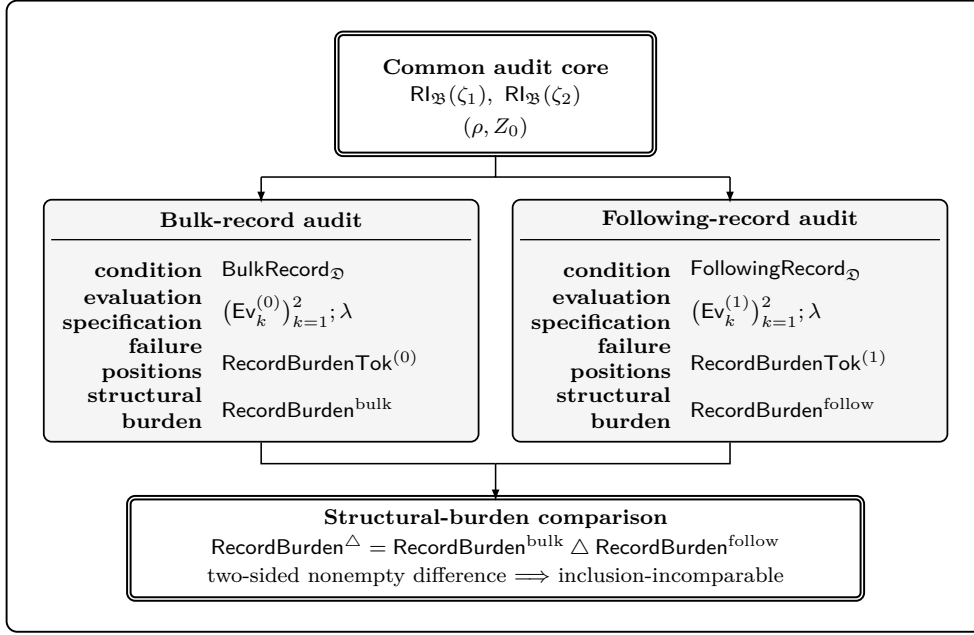


Figure 28: Configuration-specific outputs of the classical multi-object audit example. Role instances, full evaluation specifications,  $\mathfrak{D}$ -carrier conditions, record evidence,  $\mathfrak{D}$ -side evidence, and the multi-object relation are supplied as inputs, and the structural burdens relative to engineering evidence for the bulk-recording and following-recording conditions are compared.

## Collapse of the Mystery Basis

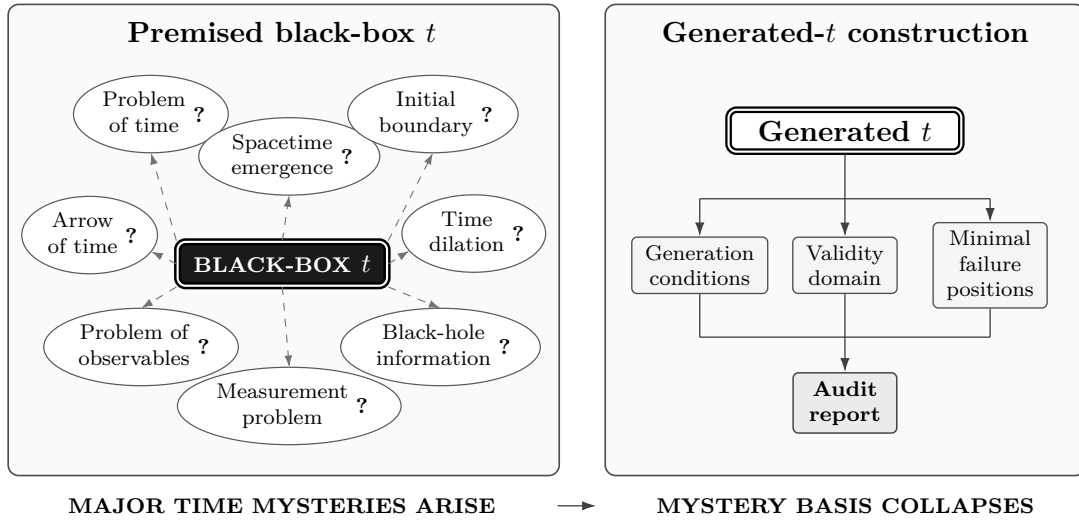


Figure 29: A time variable  $t$  presupposed as a black box without a generation path supplies the basis on which the problem of time, time stretching, initial boundaries, the arrow of time, the problem of observables, the measurement problem, the black-hole information problem, and spacetime emergence appear as mutually independent mysteries. Once time content and its grounding structure are generated from nontemporal physical inputs, the basis on which these mysteries are constituted is lost, and each problem is transferred to generation conditions, scale-indexed validity domains, the six failure series, and audit reports.

Concrete time-use audit map

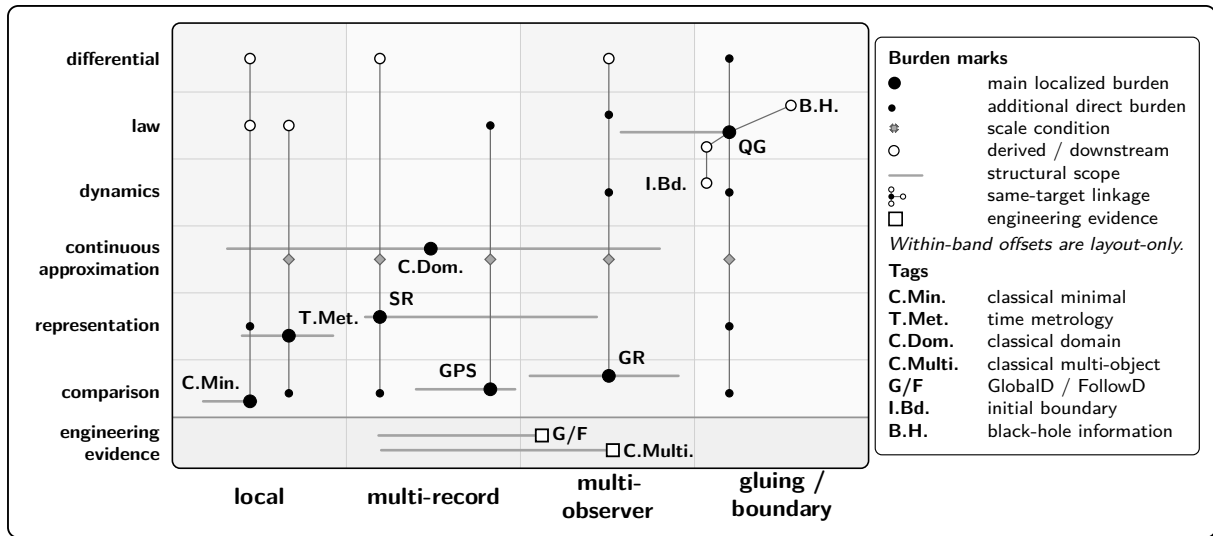


Figure 30: Correspondence between structural reach and generation-and-audit burdens in representative uses of time.

## Validity responsibility across theory and engineering / implementation

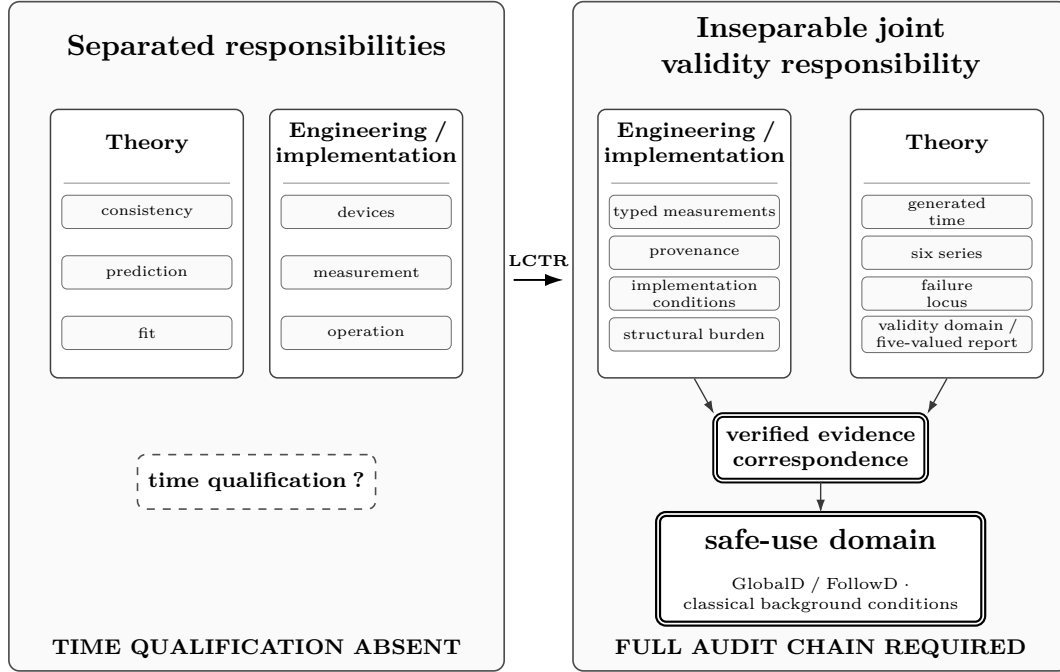


Figure 31: Connection of the final synthesis to engineering audit and classical world representations. The engineering measurements, auxiliary-audit flags, auxiliary-audit effects, and structural burdens of the engineering-audit layer are distinguished from the six failure series, invalidation positions, invalidation signatures, scale-indexed validity domains, and five-valued audit reports of the theoretical-establishment audit layer, and the two layers are connected through target components of verified evidence.

### Five audit outcomes across the 72-source corpus

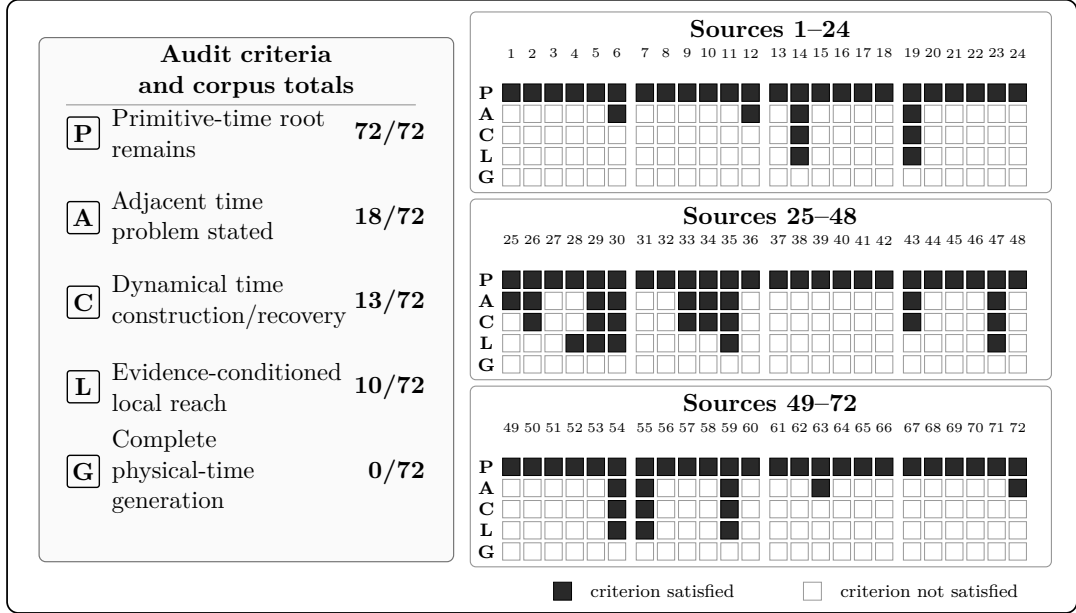


Figure 32: Source-by-source distribution of the five audit results across the 72 audited sources. Source numbers correspond to the sequence in Table 16.